Walmart Sales Forecasting & Inventory Insights

Problem Statement

A retail chain with multiple stores is facing difficulty in managing inventory due to fluctuating customer demand. Improper demand estimation leads to overstocking in some stores and stock shortages in others.

The purpose of this project is to analyze historical weekly sales data of Walmart stores to:

- Understand factors influencing sales
- Identify store-level performance differences
- Detect seasonality and trends
- Forecast future sales to support better planning.

Dataset Description

- Dataset Name: Walmart Weekly Sales Dataset
- Total Records: 6,435
- Total Features: 8

Feature Details:

- Store: Unique store identifier
- Date: Week of sales
- Weekly_Sales: Total sales for the store in that week (Target Variable)
- Holiday_Flag: Indicates whether the week includes a holiday (1 = Holiday, 0 = Non-holiday)
- Temperature: Temperature during the week
- Fuel_Price: Fuel cost in the store's region
- CPI: Consumer Price Index
- Unemployment: Unemployment rate

Objective

The main objectives of this project are:

- To analyze weekly sales behavior across different stores
- To study the impact of economic and environmental factors on sales
- To identify top-performing and underperforming stores
- To detect seasonal patterns in sales data
- To forecast weekly sales for the next 12 weeks for each store

Methodology

The project follows a structured data science workflow:

1. Data loading and inspection
2. Data preprocessing and cleaning
3. Exploratory Data Analysis (EDA)
4. Statistical relationship analysis
5. Store performance evaluation
6. Time-series forecasting

Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Scikit-learn were used for analysis and modeling.

Code: Vignesh_Collab_link

EDA & Insights

- Analyzed the distribution of weekly sales using histograms and boxplots
- Identified that sales data is right-skewed with extreme high values during certain weeks
- Observed sales spikes during holiday weeks using time-series plots
- Detected variability in sales performance across different stores

Key Insight: Holiday weeks play a significant role in increasing weekly sales.

Statistical Analysis

The following factors were analyzed to understand their relationship with weekly sales:

- **Unemployment Rate:**
    - Overall correlation with weekly sales was weak
    - Certain stores showed higher sensitivity to unemployment changes
- **Temperature:**
    - Very weak correlation with weekly sales
    - No significant impact observed
- **Consumer Price Index (CPI):**
    - Minimal correlation with weekly sales
    - CPI does not strongly influence short-term sales patterns

Conclusion: Economic indicators have limited direct impact on weekly sales at an overall level.

Forecasting Approach

- Time-series forecasting techniques were applied to predict future sales
- Store-wise sales data was ordered chronologically
- Models were built to capture historical trends and seasonality
- Sales were forecasted for the **next 12 weeks** for each store

Reason for 12-week horizon:

- Suitable for short-term planning
- Reduces uncertainty compared to long-term forecasts.

SARIMA Model Performance Summary

Based on the consolidated RMSE scores and visual inspection of the prediction plots, here is an analysis of the SARIMA model performance for each store:

Analysis:

- **Store 1 (RMSE: 81904.07):** The SARIMA model for Store 1 shows a good balance between quantitative performance and visual fit. The predictions generally follow the actual weekly sales trend and seasonality well, with reasonable error.

- **Store 2 (RMSE: 122675.37):** This model exhibits a decent fit, but there are noticeable deviations, especially during peak and trough periods, indicating that the model struggles to capture extreme fluctuations accurately. The RMSE is moderately high.
- **Store 3 (RMSE: 40089.50):** Quantitatively, Store 3 has the second-lowest RMSE. However, as noted in previous observations, the visual performance for Store 3 was deemed "not performing good." This discrepancy arises because Store 3 generally has significantly lower weekly sales volumes compared to Stores 1, 2, and 4. A lower absolute RMSE might still correspond to a visually less accurate prediction if the model smooths out fluctuations that are significant relative to that store's average sales. The model's predictions appear smoother and do not capture the sharp variations as effectively as some other models, despite the relatively low error magnitude.
- **Store 4 (RMSE: 129232.06):** This model recorded the highest RMSE among the five stores. Visually, the prediction plot for Store 4 clearly shows substantial deviations from the actual data, particularly around seasonal peaks and valleys. This indicates that the SARIMA model is the least effective for forecasting sales in Store 4.
- **Store 5 (RMSE: 22790.47):** The SARIMA model for Store 5 is the **best-performing** model. It has the lowest RMSE, and the prediction plot confirms a very strong visual fit, with predictions closely mirroring the actual weekly sales. The model successfully captures the trends and seasonal patterns for this store.

Summary of SARIMA Model Performance for Weekly Sales Forecasting

Based on the Root Mean Squared Error (RMSE) scores and the visual assessment of the prediction plots for each of the five stores, we can summarize the overall effectiveness of the SARIMA models:

Consolidated RMSE Scores:

Store	RMSE
1	81904.07
2	122675.37
3	40089.50
4	129232.06
5	22790.47

Overall Effectiveness and Observations:

1. **General Performance Trends:** The SARIMA models demonstrate varying degrees of accuracy across different stores. Stores 3 and 5 exhibit significantly lower RMSE values,

suggesting that the model performed best for these stores. Conversely, Stores 2 and 4 have the highest RMSE values, indicating poorer predictive performance.

2. **Store 1 (RMSE: 81904.07):** The SARIMA model for Store 1 shows a reasonable fit to the actual weekly sales data. The prediction plot indicates that the model generally captures the seasonal patterns and overall trend, though there are some deviations, particularly during peak sales periods.
3. **Store 2 (RMSE: 122675.37):** This store exhibits one of the higher RMSE values. Visually, the model struggles to accurately track the sharper fluctuations and magnitude of sales, especially during periods of high volatility. The predictions tend to smooth out the actual data, leading to a less precise forecast.
4. **Store 3 (RMSE: 40089.50):** Despite having the lowest RMSE score, the prediction plot for Store 3 was noted as 'not performing good' in previous analysis. This suggests a potential discrepancy between the quantitative metric and the visual fit. While the error magnitude might be small in absolute terms relative to the sales volume, the model might not be capturing the specific dynamics or turning points effectively, or there could be periods where the predictions are consistently off, but other periods balance out the overall RMSE.
5. **Store 4 (RMSE: 129232.06):** With the highest RMSE, the SARIMA model for Store 4 is the worst-performing among the five. The prediction plot likely shows significant discrepancies between forecasted and actual sales, indicating that the model struggles to capture the underlying patterns or is highly sensitive to noise in this store's data.
6. **Store 5 (RMSE: 22790.47):** This store has the second-lowest RMSE and shows good performance. The prediction plot should reflect a close alignment between the forecasted and actual sales, indicating that the SARIMA model effectively captures the trends and seasonal variations for Store 5.

Challenges and Specific Observations:

- **Seasonality:** The initial seasonal decomposition and plot analysis confirmed seasonality in all stores, which SARIMA models are designed to handle. However, the varying performance suggests that the strength and complexity of seasonality, or other uncaptured factors, differ significantly between stores.
- **Model Fit Variability:** The significant difference in RMSE scores (e.g., Store 5 vs. Store 4) highlights that a single SARIMA model configuration may not be optimal for all stores, even with similar pre-processing steps. Store-specific characteristics or external factors

not included in the model (e.g., local events, promotions) might play a larger role in some stores than others.

- **Discrepancy (RMSE vs. Visual for Store 3):** The observation for Store 3 underscores the importance of both quantitative metrics and visual inspection. A low RMSE alone doesn't always guarantee a visually 'good' fit, especially if the errors are systematic but small, or if the model misses critical turning points.

In conclusion, while SARIMA proves effective for forecasting weekly sales for some stores (e.g., Store 5 and, quantitatively, Store 3), its performance is inconsistent across the board. Further investigation into the characteristics of each store's sales data, or exploration of more complex models or external regressors, might be necessary to improve forecasting accuracy for the underperforming stores.

Summary:

Data Analysis Key Findings

- **Varying Model Accuracy**: The SARIMA models demonstrated inconsistent effectiveness across the five stores, with Root Mean Squared Error (RMSE) values ranging significantly.
- **Best Performing Models**:
  - Store 5 had the lowest RMSE at $22,790.47, indicating the most accurate predictions and a strong visual alignment between forecasts and actual sales.
  - Store 3, despite having the second-lowest RMSE at $40,089.50, showed visual discrepancies, with its model struggling to capture specific fluctuations effectively. This suggests that while its average error is low, the model might not be accurately representing critical turning points, possibly due to the store's overall lower sales volume.
- **Worst Performing Model**: Store 4 exhibited the highest RMSE at $129,232.06, indicating the least accurate predictions with significant visual deviations from actual sales.
- **Moderately Performing Models**:
  - Store 1 had a moderate RMSE of $81,904.07, demonstrating a reasonable fit that captured general trends and seasonality.
  - Store 2 had a relatively high RMSE of $122,675.37, struggling to accurately track sharper fluctuations and the magnitude of sales peaks and troughs.
- **Importance of Combined Assessment**: The analysis highlighted that relying solely on quantitative metrics like RMSE can be misleading, as demonstrated by the discrepancy

observed in Store 3 where a low RMSE did not fully align with visual assessment of the prediction quality.

Insights or Next Steps

- **Store-Specific Optimization**: Given the significant variability in performance, implementing store-specific SARIMA model parameters or exploring alternative models tailored to the unique characteristics of each store's sales data could improve forecasting accuracy.
- **Enhanced Feature Engineering**: For underperforming stores (e.g., Store 2 and Store 4), further investigation into external factors such as promotional activities, local events, holidays, or economic indicators not included in the current model could provide valuable regressors to enhance prediction capabilities.
- **Best-Performing Model:** The SARIMA model for **Store 5** demonstrated the most accurate predictions, characterized by the lowest RMSE and a highly consistent visual fit with the actual sales data.
- **Worst-Performing Model:** The SARIMA model for **Store 4** is the least effective, evidenced by the highest RMSE and significant visual discrepancies between its predictions and the actual sales.

Conclusion & Learnings

This project demonstrates an end-to-end application of data science techniques on real-world retail data.

Key Learnings:

- Performing structured EDA with business context
- Applying statistical analysis to interpret data relationships
- Understanding store-level sales behavior
- Implementing time-series forecasting for short-term prediction
- Interpreting results responsibly by acknowledging limitations

This capstone project fulfills all the requirements specified in the problem statement and reflects practical data analysis and modeling skills.